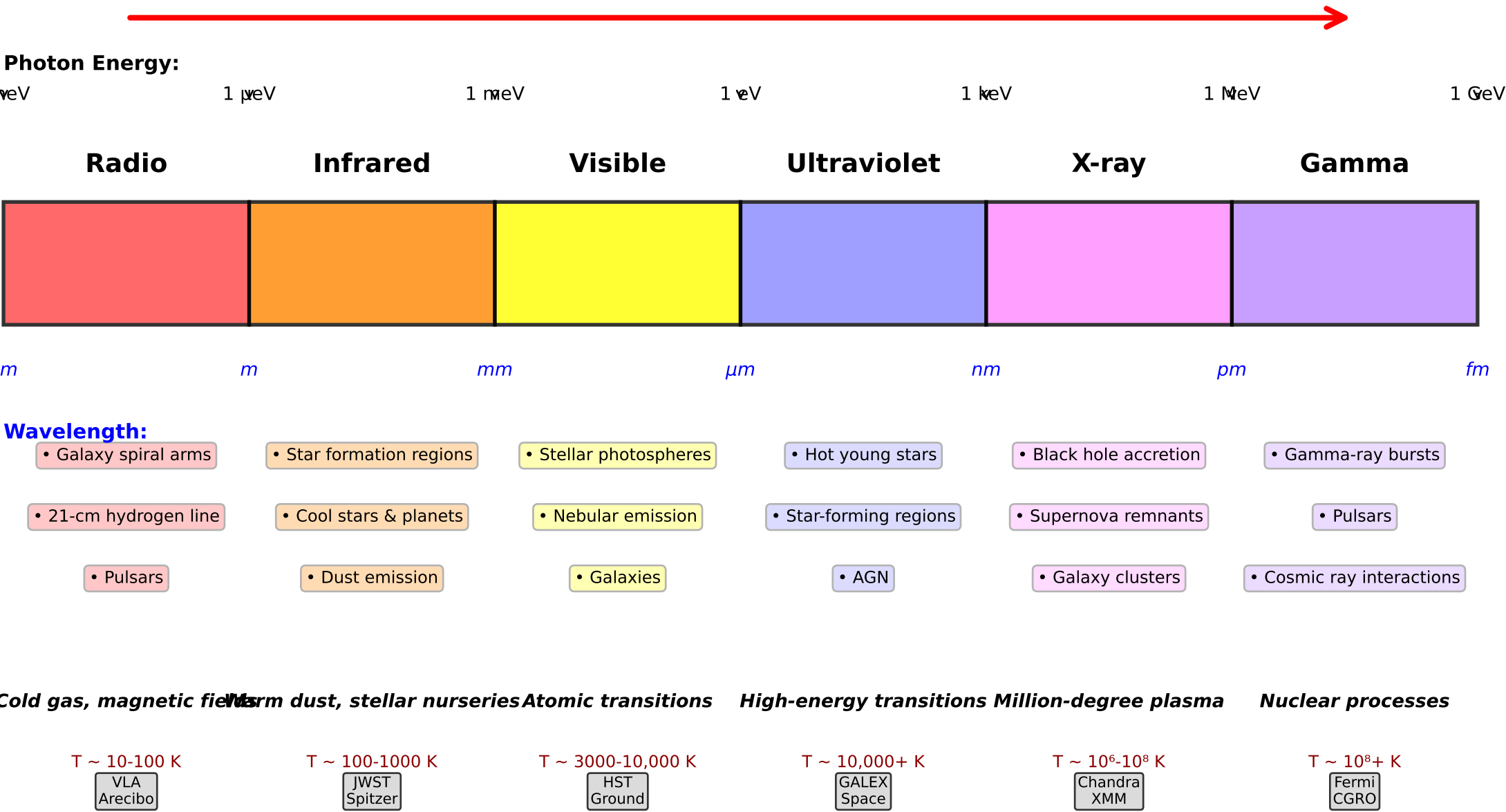


The Electromagnetic Spectrum: A Physics Ladder for Astronomy

Each wavelength band reveals different astronomical phenomena and physical processes.
Higher photon energies probe increasingly violent and exotic cosmic physics.

PHYSICS LADDER

Increasing Energy → More Violent Physics



Why Multi-Wavelength Astronomy Matters:

- Each wavelength reveals different physics
- Complete picture needs full spectrum
- Different instruments required
- Energy determines accessible processes

Wien's Law Connection:

$$\lambda_{\max} = \frac{0.29 \text{ cm}\cdot\text{K}}{T}$$

Peak wavelength $\propto 1/\text{Temperature}$